

VENKATESH VISHWANADULA

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PROFESSIONAL SUMMARY

Software Engineer with experience in building scalable full-stack web applications using the MERN stack. Proficient in React.js, Node.js, Express.js, and MongoDB, with hands-on experience in designing REST APIs, implementing authentication and authorization, optimizing frontend performance, and developing production-ready features. Passionate about writing clean, maintainable code and delivering secure, high-performance applications through effective collaboration.

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, TypeScript, React.js, Redux, Redux Toolkit, Tailwind CSS, MUI

Backend: Node.js, Express.js, MongoDB, Mongoose, MySQL, PostgreSQL, Redis, REST APIs, JWT Authentication, Google OAuth, Socket.IO

Cloud & DevOps: Docker, AWS (EC2, S3, IAM), Nginx, GitHub Actions

Testing: Jest, React Testing Library, Supertest

Tools: Git, GitHub, Postman, VS Code

EXPERIENCE

Software Engineer – SurveyHeart

10/2024 – Present

- Developed a Community Platform with public thread browsing and **authentication-based access control**, implementing protected create/edit workflows, and seamless login redirection with navigation state restoration in **React.js**.
- Built a dynamic FAQ Management System rendering **admin-managed content** across Web, iOS, and Android platforms, where each FAQ supports platform-specific sub-sections under a single question, enabling structured and reusable content delivery without code changes using **React.js** and API-driven architecture.
- Improved frontend performance using **React.memo**, **lazy loading**, **pagination**, and **debounced API calls**, reducing unnecessary re-renders and optimizing data-heavy UI rendering.
- Designed and implemented a **Multi-Tenant Architecture** subdomain platform enabling customizable landing pages with branding, testimonials, galleries, **lead capture**, and appointment booking.
- Enhanced application security by integrating backend-controlled **AWS S3 file upload workflows**, removing direct frontend uploads and strengthening secure media handling in **Node.js**.

PROJECTS

AI-Powered Study Platform

GitHub | Live

- Developed a full-stack **AI-powered study platform** using **Node.js**, **Express.js**, **MongoDB**, **React**, **TypeScript**, **Redux Toolkit**, and **Tailwind CSS**, enabling users to upload documents and generate **AI-powered summaries**, **quizzes**, **flashcards**, and **contextual chat** using **Google Gemini AI**.
- Designed and implemented scalable **RESTful APIs** following a modular **Controller-Service architecture**, with secure **JWT Authentication**, authorization middleware, protected routes, and centralized error handling.
- Designed efficient **MongoDB schemas** using **Mongoose** with schema validation, references, indexing, population, and optimized CRUD operations for documents, quizzes, flashcards, chat history, summaries, and user management.
- Built a complete **AI Quiz Engine** featuring quiz generation, answer submission, automatic score calculation, detailed result analysis, question review, progress tracking, and interactive learning workflows.

- Developed an interactive **Flashcard Learning System** with card flip animations, review tracking, starred cards, progress visualization, reusable component architecture, and efficient state management using **Redux Toolkit Async Thunks**.
- Implemented a scalable **feature-based frontend architecture** with reusable React components, centralized API communication using **Axios**, global state management through **Redux Toolkit**, responsive UI design, optimistic updates, and comprehensive loading/error handling.

EDUCATION

B.Tech – Electronics and Communication Engineering

2019 – 2023

Swarna Bharathi Institute of Science and Technology